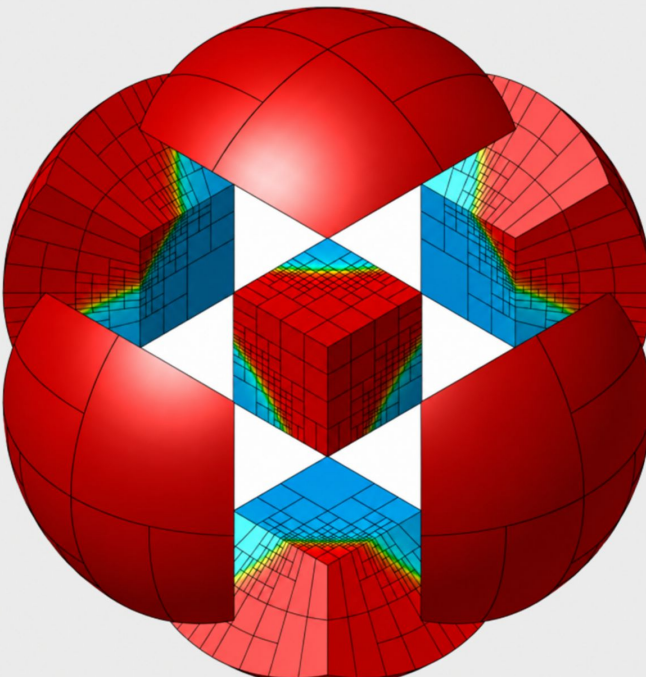




MFEM-Based Grad-Shafranov Solvers

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1. Grad-Shafranov Equation

Problem: find a static, axisymmetric plasma equilibrium:

$$\begin{aligned} \mathbf{J} \times \mathbf{B} &= \nabla p, & \nabla \cdot \mathbf{B} &= 0 \\ \mathbf{B} &= \frac{1}{r} \nabla \psi \times \hat{\mathbf{e}}_\phi + \frac{f(\psi)}{r} \hat{\mathbf{e}}_\phi, & \mathbf{J} &= \frac{1}{\mu_0} \nabla \times \mathbf{B} \\ f(\psi) &= r B_\phi, & p &= p(\psi) \end{aligned}$$

Goal: compute ψ by solving the Grad–Shafranov equation:

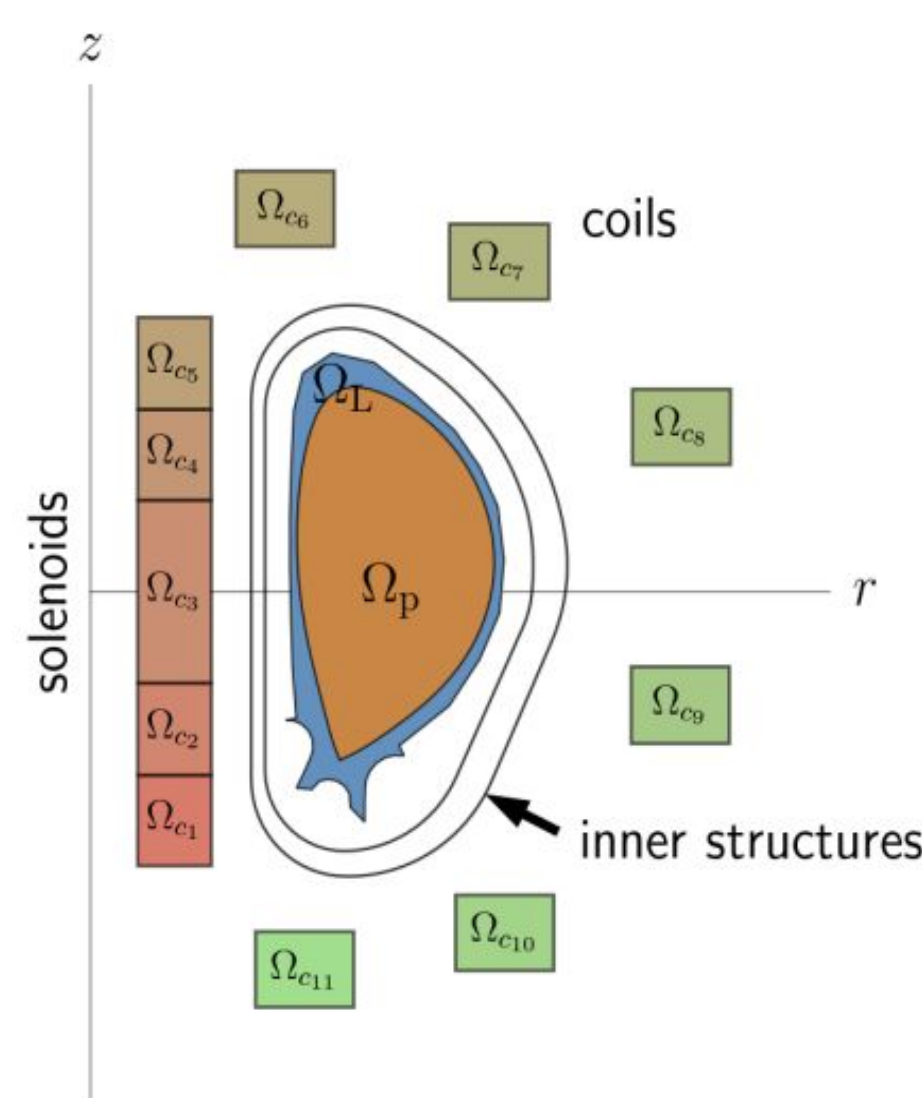
$$\Delta^* \psi = -\mu_0 r^2 p'(\psi) - f(\psi) f'(\psi)$$

Free boundary: plasma boundary determined with the equilibrium.

Fixed boundary: plasma boundary prescribed.

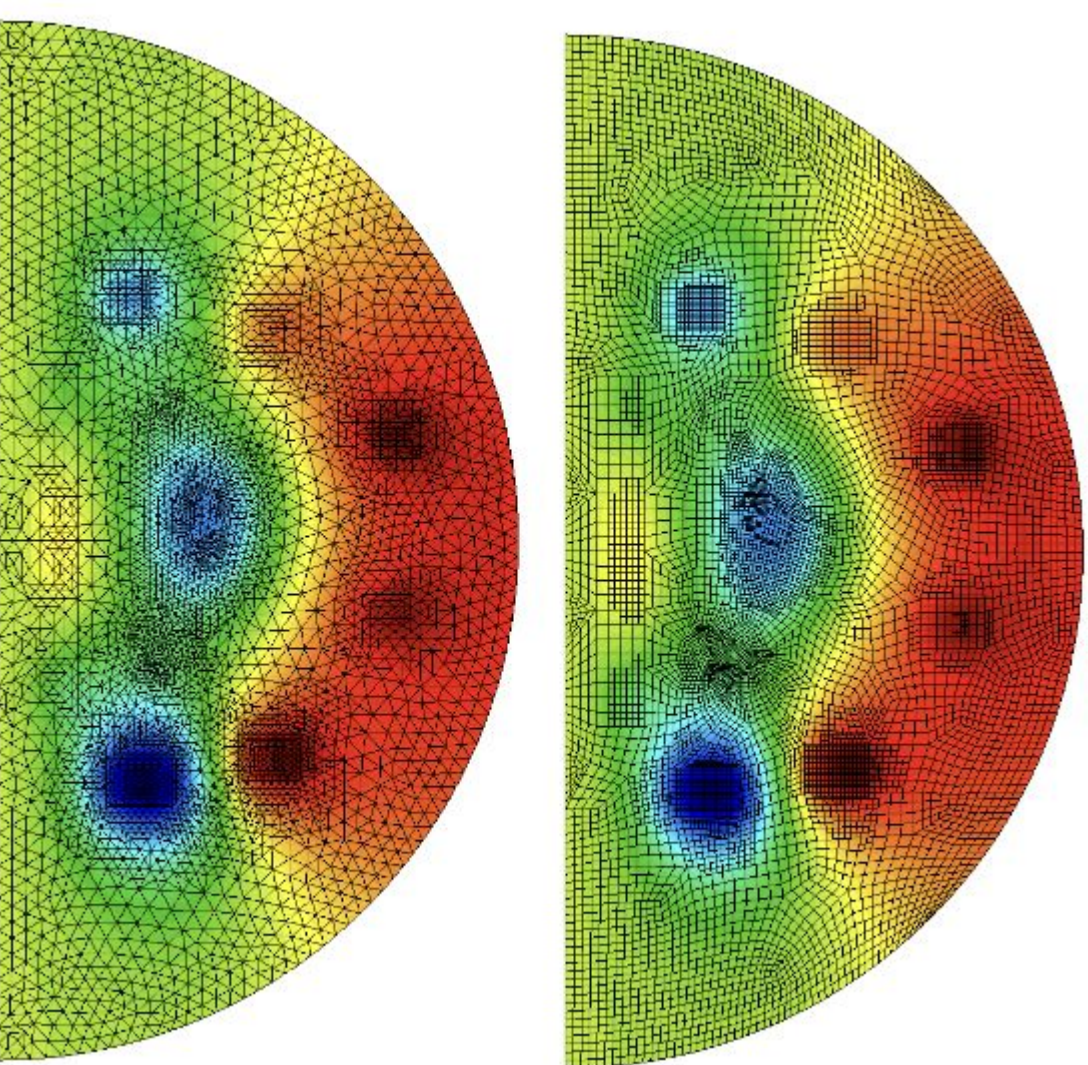
$$-\frac{1}{\mu_0 r} \Delta^* \psi = \begin{cases} r p'(\psi) + \frac{1}{\mu_0 r} f(\psi) f'(\psi), & \text{in } \Omega_p(\psi), \\ I_i / |\Omega_{c_i}|, & \text{in } \Omega_{c_i}, \\ 0, & \text{else} \end{cases}$$

$$I_p = \int_{\Omega_p(\psi)} \left(r p'(\psi) + \frac{1}{\mu_0 r} f(\psi) f'(\psi) \right) dr dz.$$



2. CG Free-Boundary GS

Background: previous work by Serino et al. (2025) established an adaptive Newton-based CG finite element Grad-Shafranov solver implemented in MFEM. Since then, some further improvements were made:



Support both triangular and quad meshes with AMR

GEQDSK export and safety factor

Interpolate ψ onto a uniform (r,z) grid and export equilibrium profiles and boundary geometry.

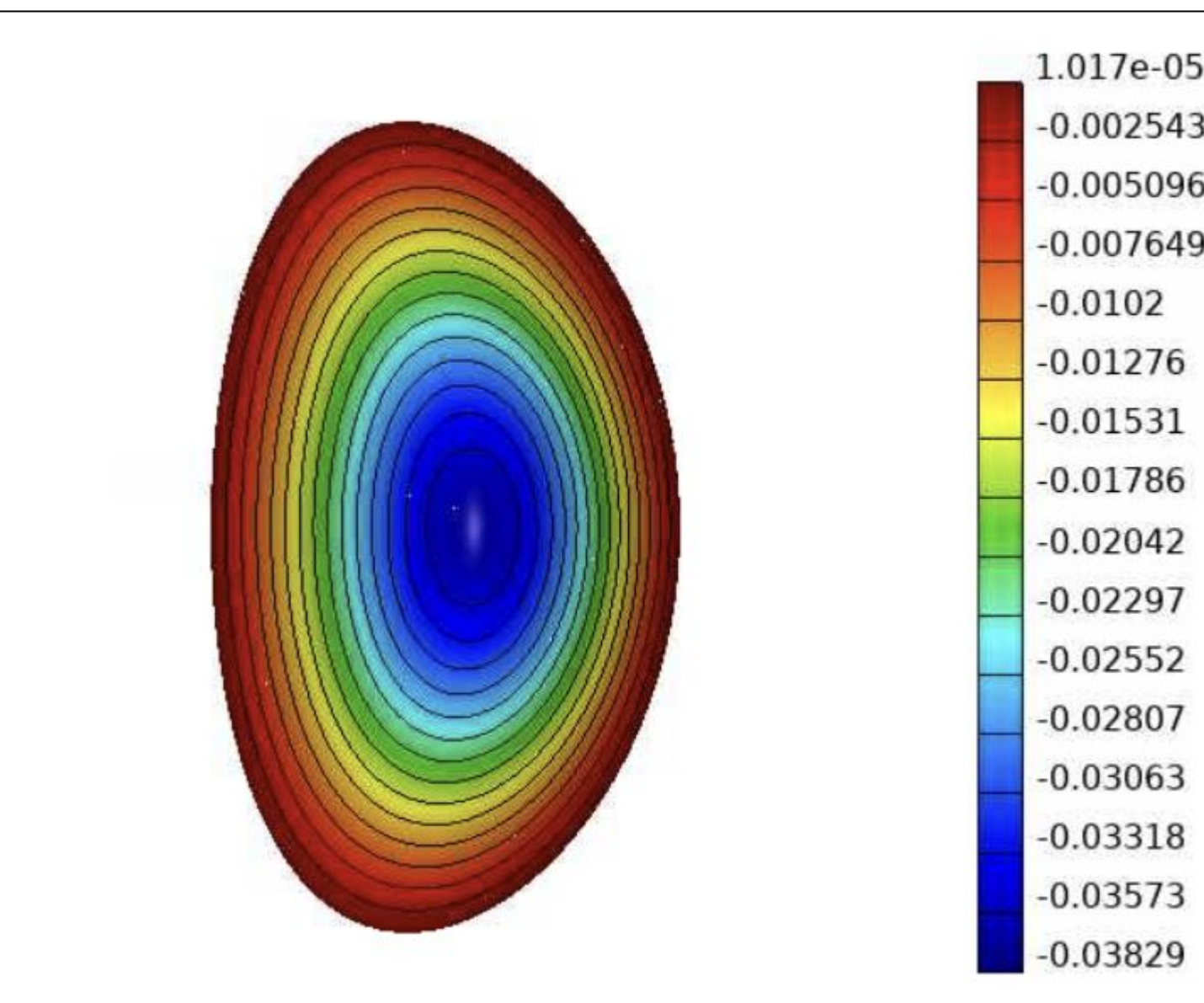
$$q(\psi) = \frac{1}{2\pi} \oint \frac{f(\psi)}{r |\nabla \psi|} dl$$

Numerical integration along closed flux contours using trapezoidal quadrature.

3. DG Fixed-Boundary Solvers

Solov'ev equilibrium

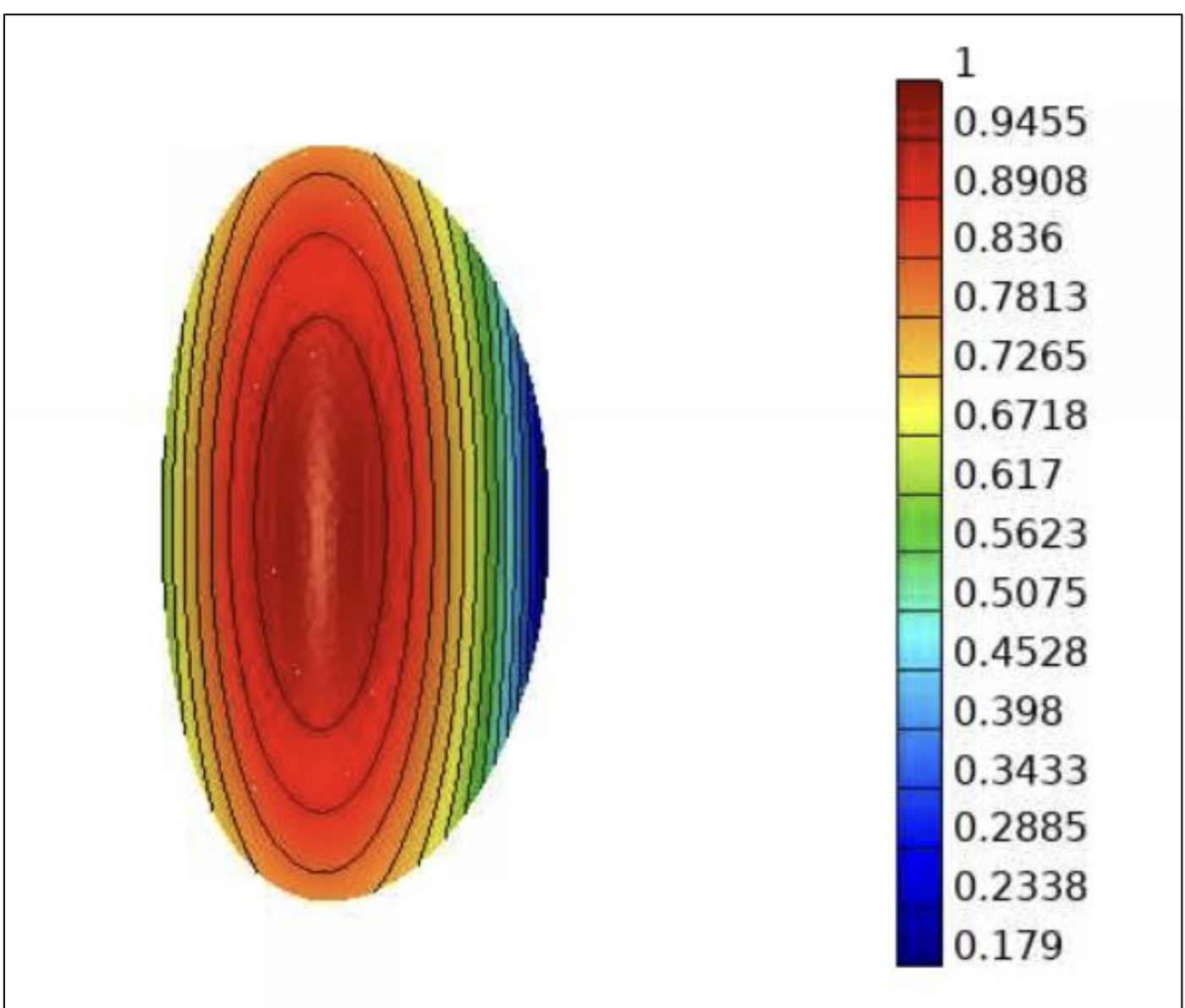
$$\begin{aligned} \Delta^* \psi &= r^2 & \text{in } \Omega_p \\ \psi &= 0 & \text{on } \partial \Omega_p \end{aligned}$$



Linear Solov'ev equilibrium

Manufactured equilibrium with nonlinear source:

$$\begin{aligned} \Delta^* \psi &= -S(r, z, \psi) & \text{in } \Omega_p \\ \psi &= \psi_{\text{ex}} & \text{on } \partial \Omega_p \end{aligned}$$



Nonlinear manufactured equilibrium

4. DG Nonlinear Solver Formulation

DG discretization: SIPG formulation, weakly enforced Dirichlet BCs

Newton linearization: explicit Jacobian construction

$$\mathcal{R}(\psi) = K_{\text{DG}} \psi - b(\psi)$$

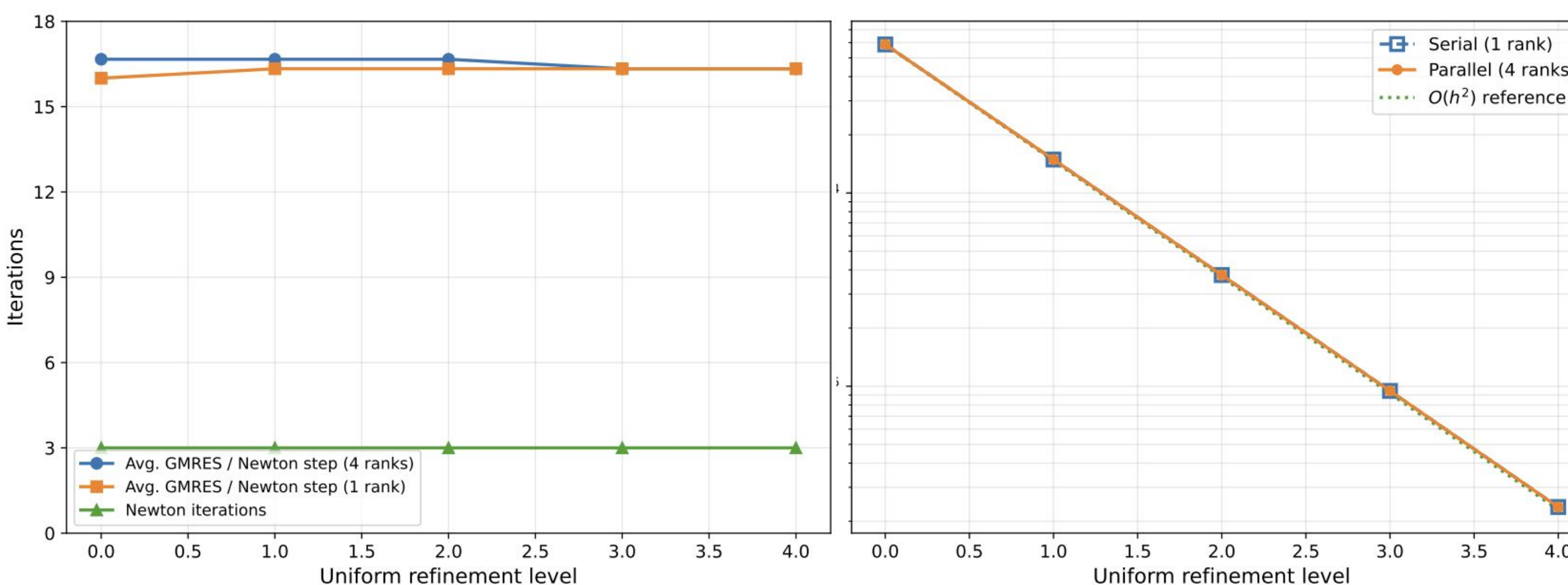
$$J(\psi_k) \delta \psi = -\mathcal{R}(\psi_k), \quad J(\psi_k) = K_{\text{DG}} - b'(\psi_k)$$

Preconditioner: Hypre BoomerAMG applied to Jacobian

Motivation: This is a first step towards a DG free-boundary Grad-Shafranov solver. With a DG representation of ψ , in a free-boundary solver, we hope to reduce force balancing errors while maintaining the divergence-free condition while transferring Grad-Shafranov equilibria to initialize 3D MHD simulations.

5. DG Solver Performance

Newton-Krylov with BoomerAMG on $J(\psi_k)$



Left: number of Newton iterations and average number of GMRES iterations per Newton iteration remain constant as problem size increases. **Right:** Solver maintains second-order convergence for P^1 DG space as problem size increases.

6. Toward DG Free-Boundary GS

Next step: free-boundary DG Grad-Shafranov solver implemented in MFEM

Established so far: CG free-boundary solver, DG fixed-boundary solver

Open questions:

- How can we detect the X- and O-points accurately inside elements in a high-order DG representation?
- Can we use TMOP to align the mesh with the separatrix? If so, is it feasible to align the mesh with the other contours of ψ ?
- Can we use CutFEM to reduce or avoid needing to better resolve the separatrix? CutFEM needs to support DG and the CutFEM integral will be part of the nonlinear system.
- How can we impose the far-field boundary condition weakly on DG?
- Can we leverage GPUs through cuDSS? Will the total plasma constraint, which leads to a less sparse matrix, be problematic?

References:

- Z. Peng, Q. Tang, and X.-Z. Tang, An adaptive discontinuous Petrov–Galerkin method for the Grad–Shafranov equation, SIAM J. Sci. Comput., 42 (2020), pp. B1227–B1249, <https://doi.org/10.1137/19M1309894>
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- R. Zhang, G. Wimmer, and Q. Tang, Structure-preserving transfer of Grad–Shafranov equilibria to magnetohydrodynamic solvers, J. Comput. Phys., 568 (2027), article 115369, <https://doi.org/10.1016/j.jcp.2026.115369>